Space-based Forest Fire Risk Assessment and Mapping: A Case Study of Wayanad District, Kerala

Abstract: Real-world disasters such as forest fires frequently result in significant losses of life, property, and ecosystems. These are all seriously threatened by forest fires, which makes risk mapping an essential tool for preventive management and mitigation. This research investigates the forest-prone areas of Wayanad district in Kerala, India. The Frequency Ratio (FR) technique is used in this work to map the prone area of forest fires using a Geographic Information System (GIS). The FR technique is an analytical tool in spatial statistics that identifies high-risk locations by correlating fire incidence with different environmental and socioeconomic parameters. Environmental factors such as vegetation type, terrain, land use, and climatic conditions were combined with data from past fire incidents to create a GIS framework. The association between these factors and the incidence of fire incidents was measured using the FR approach. The present study created a risk index that shows locations of highly prone areas of forest fires by figuring out the frequency ratio for each element. According to the study, the risk of fire is greatly influenced by elements including the density of vegetation, the slope, and the closeness to metropolitan areas. Targeted preventive and response tactics are made easier by the risk map that is produced using GIS technologies and gives a visual depiction of areas that are prone to fires. This approach helps policymakers and forest managers better allocate resources and prioritize areas for improvement. By combining the frequency ratio technique with GIS technology, forest fire risk assessment is demonstrated to be robustly approached, providing insightful information that can be used to improve overall forest management practices and lessen the effect of forest fires.

Keywords: Climatic Variables, GIS, Wayanad, Frequency Ratio, Forest Fire Risk

1. Introduction

Forests occupy around 43 million km^2 or roughly 30% of the total land area in the tropical, temperate, and boreal regions, they also help regulate the hydrologic cycle, safeguard soil resources, and serve as recreational, spiritual and aesthetic purposes for the people [1]. Therefore, it is crucial to preserve the forest's virgin ecosystem and its vegetation. Increases in the number of forest fires have resulted in a significant change in the climate [2]. Human actions comprise almost 90% of the forest fires while the remaining is contributed by lightning strikes [3]. Various studies have shown that more amounts of gases and particles such as aerosols are produced by forest fires, which play a major role in tropospheric chemistry and climate.[4]. With over 275 million inhabitants, about 30% of all Indian districts are extremely hotspots for forest fires. High-intensity Forest fire events are common in Andhra Pradesh Odisha, Maharashtra, Madhya Pradesh, Chhattisgarh, Uttarakhand, Telangana, and the other states in the northeastern region, except Sikkim**.**[5] Fire season starts from the month of March to May in the Indian subcontinent [6]. In 2009, the warmest year since 1901, there were a lot of serious forest fires. Areas of forest burned in 2010 that were small made up 44% of the total. Over 90% of the burned area was always caused by fires, which were more common among the vegetation types in the dry deciduous forest**.**[7]. Every year, fires threaten to destroy 55% of India's total forest cover, these fires are mostly caused due to burning wood fires, controlled burning, deforestation, and slash-and-burn farming methods. [8].

In Spain's Mediterranean coast, where wildfires are more common, thematic mapper data and geographic information were used to create a fire susceptibility map, with a fire in 1985 showing that over 22% of high-hazard pixels burned compared to just 3.74% of low- hazard pixels.[9]. For monitoring the daily active fires, and estimation of annual burned areas, low-resolution satellite images were used to create a fire modelling and mapping system [10]. The research conducted near the savannah and forest transition zone in the Central African Republic used multivariate regression models with remote sensing data, finding that fires in dense forests have a strong correlation with the change in land cover, while the early fires generally do not impact land cover [11]. A forest fire susceptibility model was developed using GIS and remote sensing technologies, which helped in predicting the fire-prone areas with over 80% accuracy, with the study giving importance to human activities [12]. In southern France, an expert-driven index model was developed to manage wildfire risks by using remote sensing to calculate a relative greenness index (RGRE) from NDVI data, which heavily relates ($r2 = 0.70$) with the De Mortonne aridity index **[**13].

MODIS and the DMSP-OLS data have been used to monitor the forest fires in the subtropical and temperate forests in the state of Uttaranchal in India, based on the study, several fire incidents in the state, with the majority happening at night as opposed to during the day [14]. Using GIS and remote sensing data, a forest fire susceptibility map was developed for the Bhadra Wildlife Sanctuary, Karnataka, India, comparing various factors like vegetation and slope with actual fire-impacted zones from the year 1989 to 2006 [15]. A fire susceptibility index was developed for Kolli Hills, India, to address inexact weightage assigned to forest fire categories by studying global and regional factors, with corrections made using the AHP method, considering factors like winds and forest types in creating a more reliable fire susceptibility model [16]. Using Resourcesat 1 data, forest fire distribution in Similipal Biosphere Reserve, India, was analysed from 2004 to 2013, revealing that the largest burnt area of 1014.7 Km2 occurred in 2009, with 59.6% of the reserve's vegetation experiencing fires at different frequencies and also showing 12.2 km2 area facing damage annually [17]. A study in the Kangra Himalayas, Western India, mapped wildfire occurrences by correlating fire locations with elevations, slopes, aspects, temperature, and fuel maps, key factors identified for wildfires included forest type, low altitude, high temperature, steep slopes, and human actions, 10.7% of the forest cover in the study area was categorized as very high and high forest fire-prone areas [18].

GIS technology helped in locating fire-prone areas in the Wayanad Wildlife Sanctuary by taking into account various factors that are anthropogenic and environmental, the study employed ArcGIS 9.3 and ERDAS Imagine 9.2 software to create thematic maps and a fire risk index model to classify areas into high, moderate, and low-risk zones [19]. Wildfire potential hazard areas were identified in the Thenmala forest, by using geospatial techniques. For this study, several factors are chosen, including height, slope, aspect, distance from road, settlement, and type of LU/LC [20]. The Mannarkkad forest division's fire danger zones have been attempted to be identified and mapped in this study through the use of geospatial technologies and products, based on the study, natural factors are mostly to blame for fires [21]. Fire hazard zones were identified using GIS tools and also to study the importance of the elements that caused the start of the fire, with the help of geospatial tools, a map of the fire hazard zones was created, and the weightage was made based on the AHP method, it was discovered that 71% of the fire occurred in the map's high and very high susceptibility zones, with an area under the ROC curve value of 0.795 [22]. Using datasets from Landsat-9 and sentinel 2 satellites, spectral indices were measured and calculated for the evolution of fire-impacted forest areas in Kochi region Kerala, a strong correlation was found

between forest loss and fire area in the subtropical zone of evergreen broadleaf forests (6.9%) and the zone of coniferous broadleaf forests (18.9%) [23].

The main goal of this study is to create a fire susceptibility map in the district of Wayanad by incorporating GIS techniques, by utilizing various topographic, meteorological, and man-made features. A statistical approach was done mainly to determine the impact of various factors. To evaluate the accuracy of the model a ROC curve was done to assess the prediction rate.

2. Methodology

2.1 Study area

Wayanad is situated in the northeast region of Kerala and is settled among the mountains of the Western Ghats. The district is located between latitudes 11^0 27'N and 12^0 58'N and longitudes 74^0 52'E and 76^0 07'E. Wayanad covers an area of approximately 2,132 sq. km and is bounded by the Nilgiris district of Tamil Nadu and the Mysore and Kodagu districts of Karnataka to the north and east, by Malappuram to the south, and by Kozhikode and Kannur to the west [24]. The district experiences a strong gradient in rainfall, with the primary rainfall season occurring during the southwest monsoon. Wayanad district is situated at a height between 700-2100 meters above the mean sea level (MSL). In addition to receiving over 3000 mm of rainfall from intermittent showers, the area also receives rains from both the southwest and northwest monsoons [25] (Fig. 1)

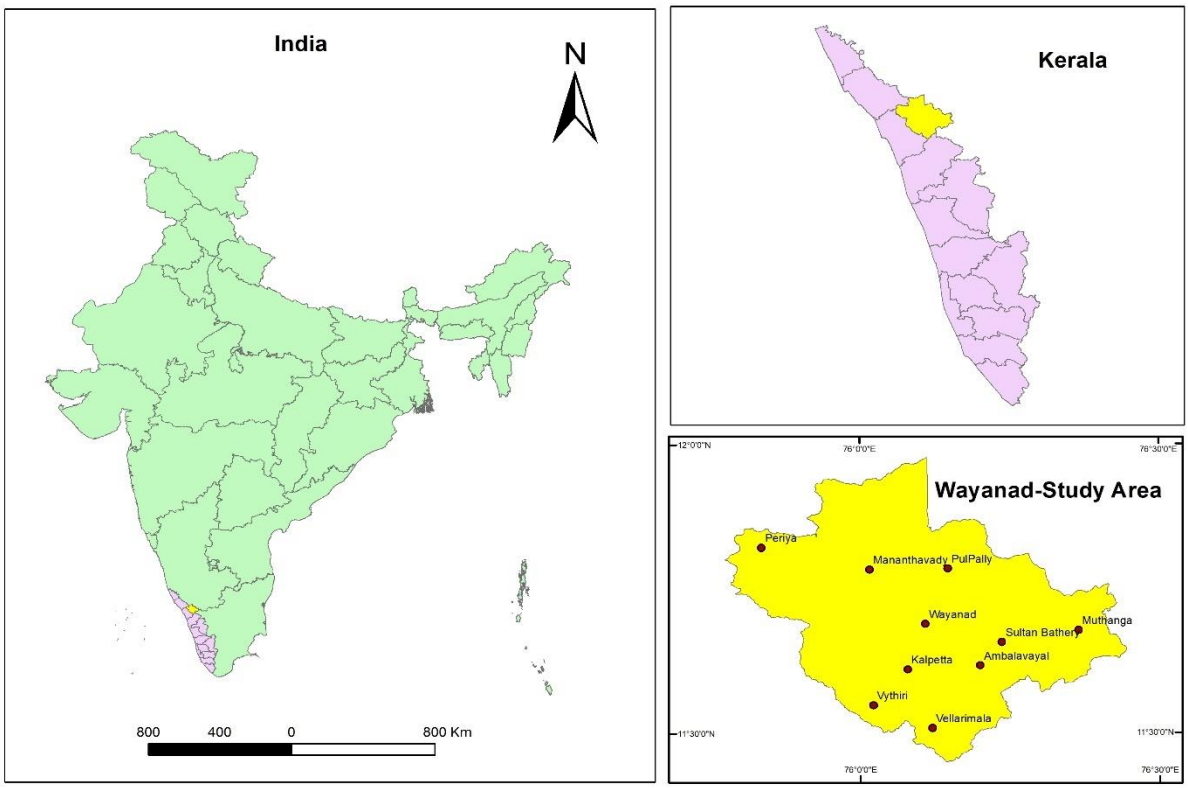


Fig. 1. Study area of the Wayanad district

2.2 Data collection

For this study, the fire points were obtained from the Forest Survey of India website. The collected points were from the duration between 2005 to 2024. About 345 fire points were obtained by using the MODIS and SNPP satellites. The Digital Elevation Model (DEM) is collected from the ISRO's Bhoonidhi website. The Slope as well as the TWI extracted from DEM. For collecting the human-related features like road networks and settlements, Open Street Maps extension and LULC data from Sentinel satellite were utilized respectively. Finally, the meteorological data were collected from the NASA Giovanni database, such as Rainfall, temperature, and wind speed for the duration of 2000 to 2024.

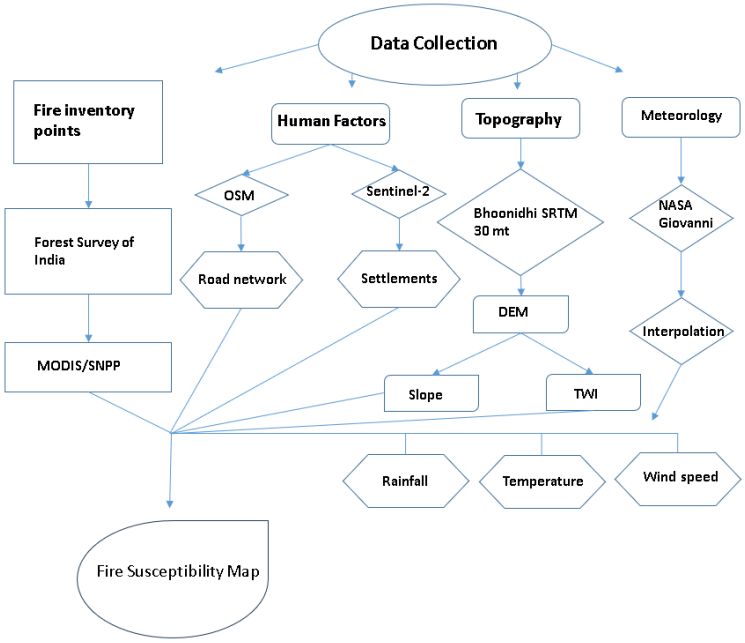


Fig. 2. Flow chart of the methodology

The connection between the risk of wildfires and the slope is complicated. While steeper slopes can make it easier for fires to spread quickly, they can also create barriers to fire movement, such as exposed rock faces or cliffs [26]. Steeper slopes were more susceptible to fires, and this was influenced by differences in land cover. However, the overall effect of slope on fire risk was consistent across all land cover types [27]. Slope data derived from the DEM was divided into five categories: 0° – 10° for gentle slopes, 10° – 20° for moderate slopes, 20° – 30° and 30° – 40° for steep slopes, and 40° – 68° for extremely steep slopes. Fires are more prone to areas that have high altitudes, therefore playing a major role in wind speed [19]. The elevation is categorized into five classes, with class 1 having the least fire susceptibility and class 5 the highest: 25 – 500m, 500 – 750m, 750 – 950m, 950 – 1250m, and 1250m>. TWI calculates the influence of topography on water movement and helps predict the spatial patterns of soil moisture and surface saturation [28]. The topographic wetness index is calculated using data from digital elevation models (DEMs). The slope angle is used as an approximation of the hydraulic gradient under steady-state conditions [29]. The topographic wetness index TWI, was grouped into five ranges: 2.4 – 5.9, 5.9 – 7.8, 7.8 – 10.7, 10.7 – 14.9, and 14.9 – 26.91. The spread of forest fires is heavily influenced by rainfall, which plays a key role in regulating fuel moisture and surface saturation levels [30]. Rainfall was split into five ranges (mm/month): 198 – 217, 182 – 198, 165 – 182, 147 – 165, 121- 147. Increased temperatures and extended periods of drought, due to climate change, are leading to more severe and frequent wildfires [31]. To better understand the relationship between weather and fires, it's important to study the temperature conditions before they occur

[32]. Temperature data is categorized into five ranges (°C): 23.01 – 23.62, 23.62 – 23.92, 23.92 – 24.20, 24.20 – 24.52, 24.52 – 24.98. High winds can fuel the rapid spread of wildfires by transporting embers and flames to new locations, allowing the fire to overcome obstacles and ignite additional areas, resulting in larger and more intense fires [33]. Wind speed (m/s): 3.6 – 3.8, 3.8 – 3.9, 3.9 – 4.1, 4.1 – 4.2, 4.2 – 4.3. The likelihood of a forest fire is higher close to roads, mostly due to the negligence and irresponsible actions of humans [19]. The proximity of roads was classified into five classes from 1 to 5, 1 being the more distance and 5 being shorter: 6000m>. 4000 – 6000m, 2000 – 4000m, 330 – 2000m, 0 – 330m. Fire is a recognized danger in informal settlements and can change urban areas. Fires can start within the informal housing structures and poor infrastructure leading to fires near settlements [34]. The proximity is divided into 5 categories: >5500m, 3500 – 5500m, 2000 – 3500m, 750 – 2000m, and 0 – 750m.

Table 1. Source and year of acquisition of data

Data/Parameters	**Source**	**Year of acquisition**
DEM 30 m	NRSC Bhuvan	-
Road Network	OpenStreetMap	-
LULC	ESRI	2023
Precipitation	NASA Giovanni	2005-2024
Temperature	NASA Giovanni	1980 - 2024
Wind speed	NASA Giovanni	2000 - 2024

2.3 Frequency Analysis

Frequency Ratio approaches identify the correlation between hotspot locations and specific factors by comparing the distribution of hotspots with the distribution of each factor in the study area [35]. The Frequency Ratio method's straightforward probability model makes it a widely used approach in research [36]. For this study, the selected parameters are classified into five classes from low to high. The parameters used in this study were classified into five categories based on established thresholds derived from published journals [37], [38], [39], [40]. These thresholds were informed by previous research that identified the relationship between parameter values and fire susceptibility. With the help of the tabulate area tool, the pixel count was calculated corresponding to the fire points collected from the FSI website.

After calculating the pixel count of each parameter, the FR (Frequency Ratio) Eq. (1), RF (Relative Frequency) Eq. (2), and PR (Predicted Ratio) Eq. (3), are derived.

FR= Total number of fire pixels / Total number of class pixels (1)

RF= FR of individual classes/ Total FR (2)

PR= Max – Min RF/ (Max-Min RF) Min RF (3)

The RF values were multiplied by hundred and then rounded to the nearest integer. These rounded values are used to reclassify the parameters again. The reclassified parameters were processed using the raster calculator tool, resulting in the creation of the fire susceptibility map.

2.4 AUC – ROC Curve

The Receiver Operating Characteristic- Area Under the Curve (ROC – AUC) analysis was conducted to evaluate the accuracy of the fire susceptibility map. The analysis involved comparing the fire inventory points with the predicted susceptibility values, providing a measure of the model's ability to distinguish between areas with and without fire occurrences. The AUC values are typically categorized into different ranges to evaluate the model's performance: 0.9 – 1.0 shows excellent model accuracy, 0.8 – 0.9 indicates good accuracy, 0.7 – 0.8 is fair accuracy, 0.6 – 0.7 exhibits poor accuracy, and 0.5 or lower depicts random or poor accuracy. Based on the value the prediction of the model can be evaluated. A higher AUC value means that the model is more reliable in predicting the onset of wildfire while a lesser value is considered as a poor or random indicator of the model.

3. Results and Discussion

3.1 Frequency Ratio

The Frequency Ratio of each parameter, the Predicted Rate (PR) was calculated. The results showed that slope and distance to roads had the highest impact on fire susceptibility, both with a PR value of 1.81. The DEM also contributed with a PR value of 1.44. other parameters, such as distance to settlements (1.29), rainfall (1.24), temperature (1.36), and TWI (1.35), had moderate influence (Table 2). Among all parameters, wind speed had the least effect on fire susceptibility, with a PR value of 1, indicating minimal impact. The analysis showed that the slope and proximity to roads had a major influence on predicting the wildfire.

Table 2. Predicted Rate of parameters

Parameter	Classes	Wildfire Pixels	FR	RF(INT)	PR
	0-10	4377	0.003	3	
	10 to 20	3535	0.006	7	
Slope	20 to 30	2712	0.014	17	
	30 to 40	1344	0.023	28	
	40 to 68	398	0.035	44	
Total		**12366**	**0.080**		**1.817734**
	25 to 500	686	0.026	32	
	500 to 750	3693	0.003	3	
DEM	750 to 950	3425	0.004	5	
	950 to 1250	3091	0.023	28	
	1250>	1471	0.024	30	
Total		**12366**	**0.080**		**1.448276**
	6000>	49	0.001	3	
	4000 to 6000	282	0.003	10	
Distance to road	2000 to 4000	1547	0.010	34	
	330 to 2000	4211	0.012	40	
	0 to 330	6277	0.004	12	
Total		**12366**	**0.030**		**1.817734**
	5500>	246	0.004	11	
	3500 to 5500	454	0.003	9	
Distance to settlements	2000 to 3500	2762	0.011	35	
	750 to 2000	4463	0.011	34	
	0 to 750	4436	0.003	9	
Total		**12361**	**0.031**		**1.295567**
	198 to 217	5098	0.010	39	
	182 to 198	2123	0.004	16	
Rainfall	165 to 182	2125	0.004	16	
	147 to 165	1853	0.004	15	
	121 to 147	1162	0.004	14	
Total		**12361**	**0.025**		**1.246305**
	23.01 to 23.62	3713	0.003	10	
	23.62 to 23.92	3303	0.006	19	
Temperature	23.92 to 24.20	1891	0.005	16	
	24.20 to 24.52	2643	0.012	38	
	24.52 to 24.98	811	0.005	15	
Total		**12361**	**0.032**		**1.364532**
	23.01 to 23.62	3545	0.007	29	
	23.62 to 23.92	4751	0.006	25	
Wind Speed	23.92 to 24.20	875	0.002	9	
`	24.20 to 24.52	1559	0.005	19	
	24.52 to 24.98	1631	0.004	14	
Total		**12361**	**0.025**		**1**
	2.45 to 5.98	72	0.002	9	
	5.98 to 7.89	394	0.002	10	
TWI	7.89 to 10.76	1305	0.004	20	
	10.76 to 14.96	4408	0.005	22	
	14.96 to 26.91	6185	0.007	36	
Total		**12364**	**0.020**		**1.35468**

3.2 Fire Susceptibility Map

The fire susceptibility map is categorized into three types namely: - Low, Moderate, and High Class. The corresponding areas for each class in square kilometres (Km^2) are as follows: Low with 822 sq. km, Moderate with 989.419 sq. km High with 300.7485 sq. km, as illustrated in (Fig. 3)

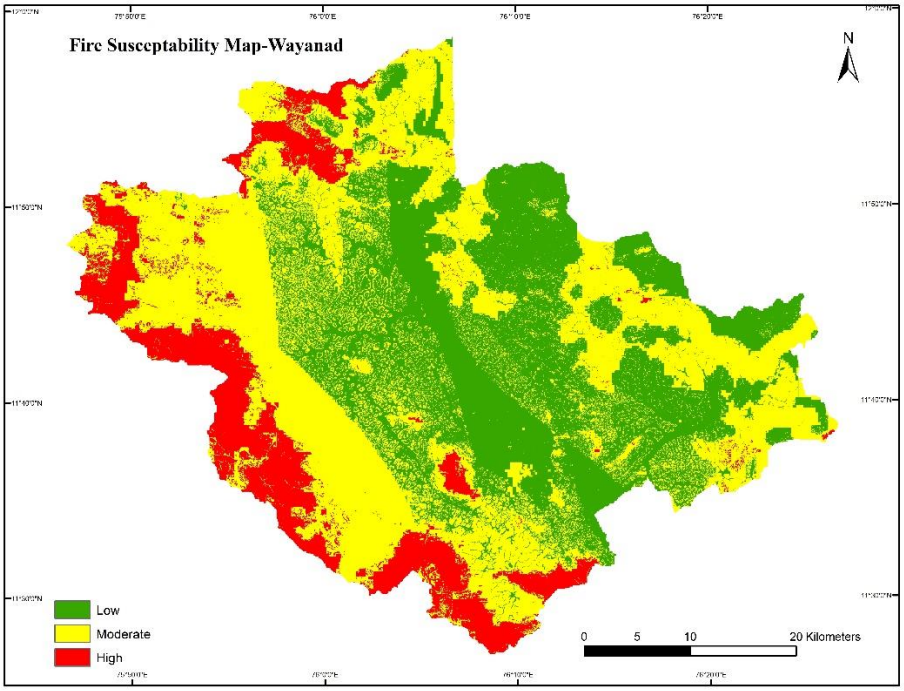


Fig. 3. Fire Susceptibility Map of Wayanad

3.3 Limitations of the study

Although a substantial number of fire points (inventory points) were obtained from the Forest Survey of India, their verification was not feasible due to the inaccessibility of reserved forest areas within the study region. The lack of field verification could impact the accuracy of the analysis. Incorporating additional verified data could have further enhanced the reliability of the results.

3.4. AUC – ROC curve

The ROC – AUC curve yielded a value of 0.733, as shown in Fig 4, indicating that the model provides a fair accuracy of fire susceptibility. By this value, it can categorize the high and low-risk zone areas with a fair or reasonable prediction.

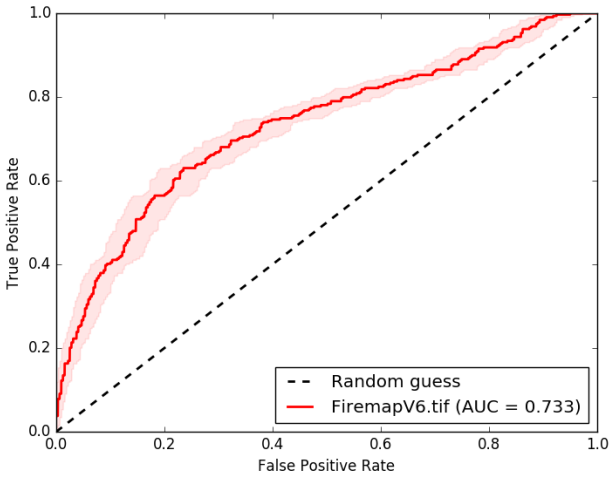


Fig. 4. ROC curve indicating Fair accuracy

4. Conclusion

In this study, a fire susceptibility map was developed using frequency analysis, incorporating various parameters such as slope, elevation, proximity to roads and settlements, and rainfall. Statistical analysis revealed that slope and proximity to roads had the greatest influence on wildfire occurrence. The resulting map serves as a valuable resource for government agencies, providing a reference or framework to guide efforts in wildfire prevention and mitigation. Several strategies can be implemented to mitigate wildfire risks in high susceptibility areas which includes the removal of dry vegetation to reduce the availability of fuel, the establishment of fire breaks in high-risk zones to limit fire spread, and the promotion of awareness programs to educate local communities about fire prevention and safety measures. Additionally, the use of fire-resistant plant species in these areas and help minimize the spread and impact of wildfires. The model was validated using real-world data by comparing the predicted fire susceptibility map with actual fire occurrence points obtained from the Forest Survey of India. To assess the model's accuracy, the Area Under the Curve (AUC) value was calculated, resulting in an AUC of 0.733. The value indicates a satisfactory level of accuracy, suggesting that the model is capable of distinguishing between high and low susceptibility areas with reasonable reliability. Additionally, the spatial distribution of high-susceptibility zones identified by the model was compared with the actual fire occurrence points, showing a significant overlap.

5. References

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